Additive synthesis

Additive synthesis is a sound synthesis technique that creates timbre by adding sine waves together.[1][2]

The timbre of musical instruments can be considered in the light of Fourier theory to consist of multiple harmonic or inharmonic *partials* or overtones. Each partial is a sine wave of different frequency and amplitude that swells and decays over time due to modulation from an ADSR envelope or low frequency oscillator.

Additive synthesis most directly generates sound by adding the output of multiple sine wave generators. Alternative implementations may use pre-computed wavetables or the inverse fast Fourier transform.

Additive synthesis example

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A bell-like sound generated by additive synthesis of 21 inharmonic partials

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Explanation

The sounds that are heard in everyday life are not characterized by a single frequency. Instead, they consist of a sum of pure sine frequencies, each one at a different amplitude. When humans hear these frequencies simultaneously, we can recognize the sound. This is true for both "non-musical" sounds (e.g. water splashing, leaves rustling, etc.) and for "musical sounds" (e.g. a piano note, a bird's tweet, etc.). This set of parameters (frequencies, their relative amplitudes, and how the relative amplitudes change over time) are encapsulated by the *timbre* of the sound. Fourier analysis is the technique that is used to determine these exact timbre parameters from an overall sound signal; conversely, the resulting set of frequencies and amplitudes is called the Fourier series of the original sound signal.

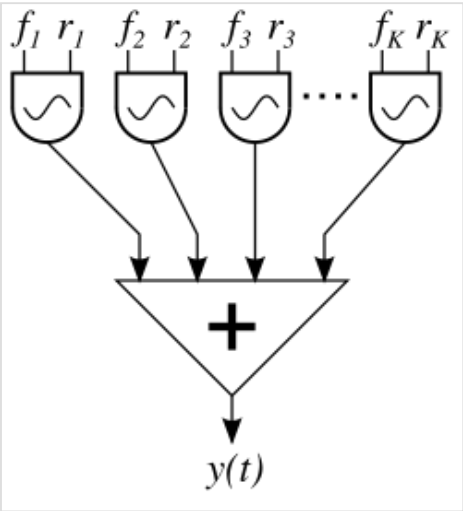
In the case of a musical note, the lowest frequency of its timbre is designated as the sound's fundamental frequency. For simplicity, we often say that the note is playing at that fundamental frequency (e.g. "middle C is 261.6 Hz"),[3] even though the sound of that note consists of many other frequencies as well. The set of the remaining frequencies is called the overtones (or the harmonics, if their frequencies are integer multiples of the fundamental frequency) of the sound.[4] In other words, the fundamental frequency alone is responsible for the pitch of the note, while the overtones define the timbre of the sound. The overtones of a piano playing middle C will be quite different from the overtones of a violin playing the same note; that's what allows us to differentiate the sounds of the two instruments. There are even subtle differences in timbre between different versions of the same instrument (for example, an upright piano vs. a grand piano).

Additive synthesis aims to exploit this property of sound in order to construct timbre from the ground up. By adding together pure frequencies (sine waves) of varying frequencies and amplitudes, we can precisely define the timbre of the sound that we want to create.

Definitions

Harmonic additive synthesis is closely related to the concept of a Fourier series which is a way of expressing a periodic function as the sum of sinusoidal functions with frequencies equal to integer multiples of a common fundamental frequency. These sinusoids are called harmonics, overtones, or generally, partials. In general, a Fourier series contains an infinite number of sinusoidal components, with no upper limit to the frequency of the

sinusoidal functions and includes a DC component (one with frequency of 0 Hz). Frequencies outside of the human audible range can be omitted in additive synthesis. As a result, only a finite number of sinusoidal terms with frequencies that lie within the audible range are modeled in additive synthesis.



Schematic diagram of additive synthesis. The inputs to the oscillators are frequencies f_k and amplitudes r_k.

A waveform or function is said to be periodic if

$$y(t) = y(t + P)$$

for all t and for some period P.

The Fourier series of a periodic function is mathematically expressed as:

$$\begin{aligned} y(t) &= \frac{a_0}{2} + \sum_{k=1}^{\infty} \left[a_k \cos(2\pi k f_0 t) - b_k \sin(2\pi k f_0 t)\right] \\ &= \frac{a_0}{2} + \sum_{k=1}^{\infty} r_k \cos(2\pi k f_0 t + \phi_k) \end{aligned}$$

where

- $f_0 = 1/P$ is the fundamental frequency of the waveform and is equal to the reciprocal of the period,
- $a_k = r_k \cos(\phi_k) = 2 f_0 \int_0^P y(t) \cos(2\pi k f_0 t)\, dt, \quad k \geq 0$
- $b_k = r_k \sin(\phi_k) = -2 f_0 \int_0^P y(t) \sin(2\pi k f_0 t)\, dt, \quad k \geq 1$
- $r_k = \sqrt{a_k^2 + b_k^2}$ is the amplitude of the kth harmonic,
- $\phi_k = \operatorname{atan2}(b_k, a_k)$ is the phase offset of the kth harmonic. atan2 is the four-quadrant arctangent function,

Being inaudible, the DC component, $a_0/2$, and all components with frequencies higher than some finite limit, $K f_0$, are omitted in the following expressions of additive synthesis.

Harmonic form

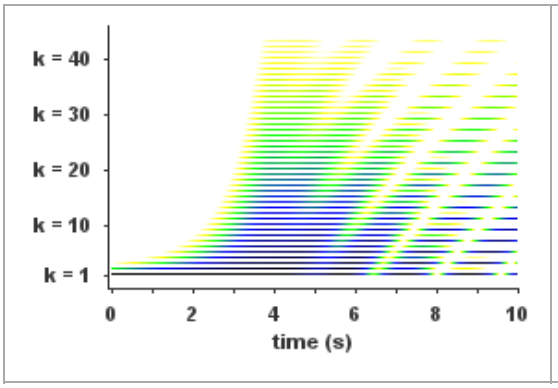
The simplest harmonic additive synthesis can be mathematically expressed as:

$$y(t) = \sum_{k=1}^{K} r_k \cos(2\pi k f_0 t + \phi_k), \tag{1}$$

where $y(t)$ is the synthesis output, r_k, $k f_0$, and ϕ_k are the amplitude, frequency, and the phase offset, respectively, of the kth harmonic partial of a total of K harmonic partials, and f_0 is the fundamental frequency of the waveform and the frequency of the musical note.

Time-dependent amplitudes

More generally, the amplitude of each harmonic can be prescribed as a function of time, $r_k(t)$, in which case the synthesis output is



Example of harmonic additive synthesis in which each harmonic has a time-dependent amplitude. The fundamental frequency is 440 Hz.

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$$y(t) = \sum_{k=1}^{K} r_k(t) \cos(2\pi k f_0 t + \phi_k). \tag{2}$$

Each envelope $r_k(t)$ should vary slowly relative to the frequency spacing between adjacent sinusoids. The bandwidth of $r_k(t)$ should be significantly less than f_0.

Inharmonic form

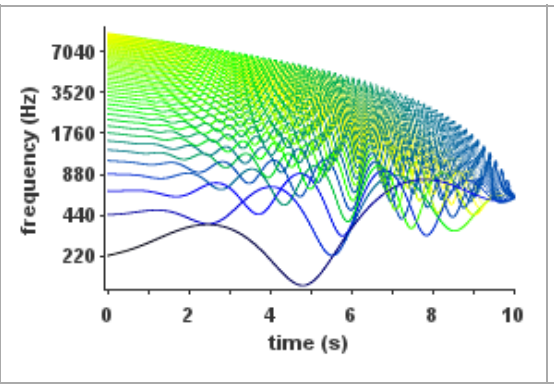
Additive synthesis can also produce inharmonic sounds (which are aperiodic waveforms) in which the individual overtones need not have frequencies that are integer multiples of some common fundamental frequency.[5][6] While many conventional musical instruments have harmonic partials (e.g. an oboe), some have inharmonic partials (e.g. bells). Inharmonic additive synthesis can be described as

$$y(t) = \sum_{k=1}^{K} r_k(t) \cos(2\pi f_k t + \phi_k),$$

where f_k is the constant frequency of kth partial.

Time-dependent frequencies

In the general case, the instantaneous frequency of a sinusoid is the derivative (with respect to time) of the argument of the sine or cosine function. If this frequency is represented in hertz, rather than in angular frequency form, then this derivative is divided by 2π. This is the case whether the partial is harmonic or inharmonic and whether its frequency is constant or time-varying.



Example of inharmonic additive synthesis in which both the amplitude and frequency of each partial are time-dependent.

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In the most general form, the frequency of each non-harmonic partial is a non-negative function of time, $f_k(t)$, yielding

$$y(t) = \sum_{k=1}^{K} r_k(t) \cos\left(2\pi \int_0^t f_k(u)\, du + \phi_k\right). \tag{3}$$

Broader definitions

Additive synthesis more broadly may mean sound synthesis techniques that sum simple elements to create more complex timbres, even when the elements are not sine waves.[7][8] For example, F. Richard Moore listed additive synthesis as one of the "four basic categories" of sound synthesis alongside subtractive synthesis, nonlinear synthesis, and physical modeling.[8] In this broad sense, pipe organs, which also have pipes producing non-sinusoidal waveforms, can be considered as a variant form of additive synthesizers. Summation of principal components and Walsh functions have also been classified as additive synthesis.[9]

Implementation methods

Modern-day implementations of additive synthesis are mainly digital. (See section *Discrete-time equations* for the underlying discrete-time theory)

Oscillator bank synthesis

Additive synthesis can be implemented using a bank of sinusoidal oscillators, one for each partial.[1]

Wavetable synthesis

In the case of harmonic, quasi-periodic musical tones, wavetable synthesis can be as general as time-varying additive synthesis, but requires less computation during synthesis.[10][11] As a result, an efficient implementation of time-varying additive synthesis of harmonic tones can be accomplished by use of *wavetable synthesis*.

Group additive synthesis

Group additive synthesis[12][13][14] is a method to group partials into harmonic groups (having different fundamental frequencies) and synthesize each group separately with *wavetable synthesis* before mixing the results.

Inverse FFT synthesis

An inverse fast Fourier transform can be used to efficiently synthesize frequencies that evenly divide the transform period or "frame". By careful consideration of the DFT frequency-domain representation it is also possible to efficiently synthesize sinusoids of arbitrary frequencies using a series of overlapping frames and the inverse fast Fourier transform.[15]

Additive analysis/resynthesis

It is possible to analyze the frequency components of a recorded sound giving a "sum of sinusoids" representation. This representation can be re-synthesized using additive synthesis. One method of decomposing a sound into time varying sinusoidal partials is short-time Fourier transform (STFT)-based McAulay-Quatieri Analysis.[17][18]

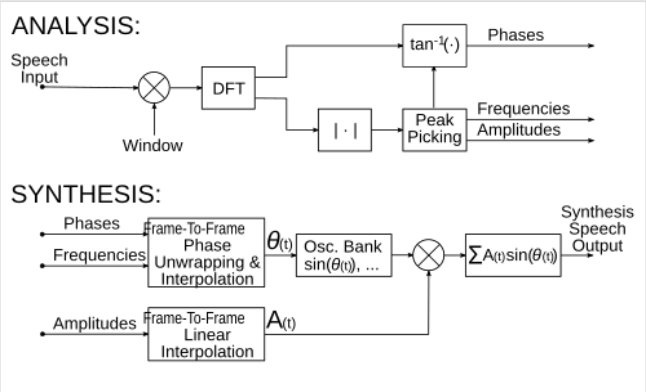
By modifying the sum of sinusoids representation, timbral alterations can be made prior to resynthesis. For example, a harmonic sound could be restructured to sound inharmonic, and vice versa. Sound hybridisation or "morphing" has been implemented by additive resynthesis.[19]

Additive analysis/resynthesis has been employed in a number of techniques including Sinusoidal Modelling,[20] Spectral Modelling Synthesis (SMS),[19] and the Reassigned Bandwidth-Enhanced Additive Sound Model.[21] Software that implements additive analysis/resynthesis includes: SPEAR,[22] LEMUR, LORIS,[23] SMSTools,[24] ARSS.[25]

Products

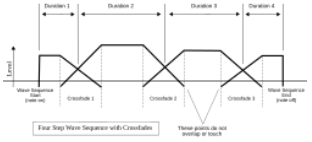
New England Digital Synclavier had a resynthesis feature where samples could be analyzed and converted into "timbre frames" which were part of its additive synthesis engine. Technos Acxel, launched in 1987, utilized the additive analysis/resynthesis model, in an FFT implementation.

Also a vocal synthesizer, Vocaloid have been implemented on the basis of additive analysis/resynthesis: its spectral voice model called Excitation plus Resonances (EpR) model[26][27] is extended based on Spectral Modeling Synthesis (SMS), and its diphone concatenative synthesis is processed using *spectral peak processing* (SPP)[28] technique similar to modified phase-locked vocoder[29] (an improved phase vocoder for formant processing).[30] Using these techniques, spectral components (*formants*) consisting of purely harmonic partials can be appropriately transformed into desired form for sound modeling, and sequence of short samples (*diphones* or *phonemes*) constituting desired phrase, can be smoothly connected by interpolating matched partials and formant peaks, respectively, in the inserted transition region between different samples. (See also Dynamic timbres)

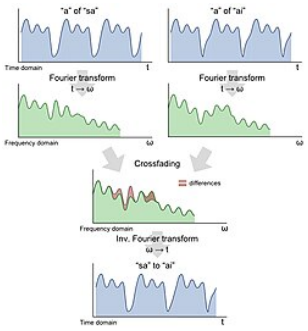


Sinusoidal analysis/synthesis system for Sinusoidal Modeling (based on McAulay & Quatieri 1988, p. 161)[16]

Additive re-synthesis using timbre-frame concatenation:



Concatenation with crossfades (on Synclavier)



Concatenation with spectral envelope interpolation (on Vocaloid)

Applications

Musical instruments

Additive synthesis is used in electronic musical instruments. It is the principal sound generation technique used by Eminent organs.

Speech synthesis

In linguistics research, harmonic additive synthesis was used in the 1950s to play back modified and synthetic speech spectrograms.[31]

Later, in the early 1980s, listening tests were carried out on synthetic speech stripped of acoustic cues to assess their significance. Time-varying formant frequencies and amplitudes derived by linear predictive coding were synthesized additively as pure tone whistles. This method is called sinewave synthesis.[32][33] Also the composite sinusoidal modeling (CSM)[34][35] used on a singing speech synthesis feature on the Yamaha CX5M (1984), is known to use a similar approach which was independently developed during 1966–1979.[36][37] These methods are characterized by extraction and recomposition of a set of significant spectral peaks corresponding to the several resonance modes occurring in the oral cavity and nasal cavity, in a viewpoint of acoustics. This principle was also utilized on a physical modeling synthesis method, called modal synthesis.[38][39][40][41]

History

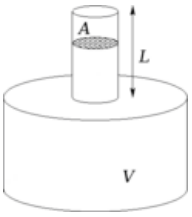
Harmonic analysis was discovered by Joseph Fourier,[42] who published an extensive treatise of his research in the context of heat transfer in 1822.[43] The theory found an early application in prediction of tides. Around 1876,[44] William Thomson (later ennobled as Lord Kelvin) constructed a mechanical tide predictor. It consisted of a *harmonic analyzer* and a *harmonic synthesizer*, as they were called already in the 19th century.[45][46] The analysis of tide measurements was done using James Thomson's *integrating machine*. The resulting Fourier coefficients were input into the synthesizer, which then used a system of cords and pulleys to generate and sum harmonic sinusoidal partials for prediction of future tides. In 1910, a similar machine was built for the analysis of periodic waveforms of sound.[47] The synthesizer drew a graph of the combination waveform, which was used chiefly for visual validation of the analysis.[47]

Lord Kelvin's Tide-predicting machine

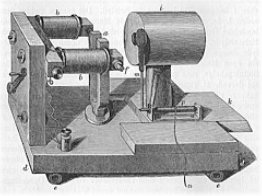
Harmonic synthesizer

Harmonic analyzer

Georg Ohm applied Fourier's theory to sound in 1843. The line of work was greatly advanced by Hermann von Helmholtz, who published his eight years worth of research in 1863.[48] Helmholtz believed that the psychological perception of tone color is subject to learning, while hearing in the sensory sense is purely physiological.[49] He supported the idea that perception of sound derives from signals from nerve cells of the basilar membrane and that the elastic appendages of these cells are sympathetically vibrated by pure sinusoidal tones of appropriate frequencies.[47] Helmholtz agreed with the finding of Ernst Chladni from 1787 that certain sound sources have inharmonic vibration modes.[49]



Helmholtz resonator



Tone-generator utilizing it

In Helmholtz's time, electronic amplification was unavailable. For synthesis of tones with harmonic partials, Helmholtz built an electrically excited array of tuning forks and acoustic resonance chambers that allowed adjustment of the amplitudes of the partials.[50] Built at least as early as in 1862,[50] these were in turn refined by Rudolph Koenig, who demonstrated his own setup in 1872.[50] For harmonic synthesis, Koenig also built a large apparatus based on his *wave siren*. It was pneumatic and utilized cut-out tonewheels, and was criticized for low purity of its partial tones.[44] Also tibia pipes of pipe organs have nearly sinusoidal waveforms and can be combined in the manner of additive synthesis.[44]

Rudolph Koenig's sound analyzer and synthesizer

sound synthesizer

sound analyzer

In 1938, with significant new supporting evidence,[51] it was reported on the pages of Popular Science Monthly that the human vocal cords function like a fire siren to produce a harmonic-rich tone, which is then filtered by the vocal tract to produce different vowel tones.[52] By the time, the additive Hammond organ was already on market. Most early electronic organ makers thought it too expensive to manufacture the plurality of oscillators required by additive organs, and began instead to build subtractive ones.[53] In a 1940 Institute of Radio Engineers meeting, the head field engineer of Hammond elaborated on the company's new *Novachord* as having a *"subtractive system"* in contrast to the original Hammond organ in which *"the final tones were built up by combining sound waves"*.[54]

Alan Douglas used the qualifiers *additive* and *subtractive* to describe different types of electronic organs in a 1948 paper presented to the Royal Musical Association.[55] The contemporary wording *additive synthesis* and *subtractive synthesis* can be found in his 1957 book *The electrical production of music*, in which he categorically lists three methods of forming of musical tone-colours, in sections titled *Additive synthesis*, *Subtractive synthesis*, and *Other forms of combinations*.[56]

A typical modern additive synthesizer produces its output as an electrical, analog signal, or as digital audio, such as in the case of software synthesizers, which became popular around year 2000.[57]

Timeline

The following is a timeline of historically and technologically notable analog and digital synthesizers and devices implementing additive synthesis.

Research implementation or publication	Commercially available	Company or institution	Synthesizer or synthesis device	Description	Audio samples
1900[58]	1906[58]	New England Electric Music Company	Telharmonium	The first polyphonic, touch-sensitive music synthesizer.[59] Implemented sinuosoidal additive synthesis using tonewheels and alternators. Invented by Thaddeus Cahill.	*no known recordings*[58]
1933[60]	1935[60]	Hammond Organ Company	Hammond Organ	An electronic additive synthesizer that was commercially more successful than Telharmonium.[59] Implemented sinusoidal additive synthesis using tonewheels and magnetic pickups. Invented by Laurens Hammond.	Model A
1950 or earlier[31]		Haskins Laboratories	Pattern Playback	A speech synthesis system that controlled amplitudes of harmonic partials by a spectrogram that was either hand-drawn or an analysis result. The partials were generated by a multi-track optical tonewheel.[31]	samples (http://www.haskins.yale.edu/featured/sentences/ppsentences.html) Archived (https://web.archive.org/web/20120125203941/http://www.haskins.yale.edu/featured/sentences/ppsentences.html) 25 January 2012 at the Wayback Machine
1958[61]			ANS	An additive synthesizer[62] that played microtonal spectrogram-like scores using multiple multi-track optical tonewheels. Invented by Evgeny Murzin. A similar instrument that utilized electronic oscillators, the *Oscillator Bank*, and its input device *Spectrogram* were realized by Hugh Le Caine in 1959.[63][64]	1964 model
1963[65]		MIT		An off-line system for digital spectral analysis and	

				resynthesis of the attack and steady-state portions of musical instrument timbres by David Luce.[65]	
1964[66]		University of Illinois	Harmonic Tone Generator	An electronic, harmonic additive synthesis system invented by James Beauchamp.[66][67]	samples (https://web.archive.org/web/20131228061841/http://ems.music.uiuc.edu/beaucham/htg_sounds/) (info (https://web.archive.org/web/20120322191551/http://ems.music.uiuc.edu/beaucham/htg.html))
1974 or earlier[68][69]	1974[68][69]	RMI	Harmonic Synthesizer	The first synthesizer product that implemented additive[70] synthesis using digital oscillators.[68][69] The synthesizer also had a time-varying analog filter.[68] RMI was a subsidiary of Allen Organ Company, which had released the first commercial digital church organ, the *Allen Computer Organ*, in 1971, using digital technology developed by North American Rockwell.[71]	1 (https://soundcloud.com/doombient-music/rmi-harmonic-drones) 2 (https://soundcloud.com/doombient-music/rmi-harmonic-demos) 3 (https://soundcloud.com/doombient-music/rmi-harmonic-arpeggiator-demo) 4 (https://soundcloud.com/doombient-music/rmi-harmonic-intermodulation)
1974[72]		EMS (London)	Digital Oscillator Bank	A bank of digital oscillators with arbitrary waveforms, individual frequency and amplitude controls,[73] intended for use in analysis-resynthesis with the digital *Analysing Filter Bank* (AFB) also constructed at EMS.[72][73] Also known as: *DOB*.	in The New Sound of Music[74]
1976[75]	1976[76]	Fairlight	Qasar M8	An all-digital synthesizer that used the fast Fourier transform[77] to create samples from interactively drawn amplitude envelopes of harmonics.[78]	samples (http://anerd.com/fairlight/audioarchives/index.htm)

1977[79]		Bell Labs	Digital Synthesizer	A real-time, digital additive synthesizer[79] that has been called the first true digital synthesizer.[80] Also known as: *Alles Machine*, *Alice*.	sample (http://retiary.org/ls/music/realaudio/ob_sys/05_alles_synth_improv.rm) Archived (https://web.archive.org/web/20110726063422/http://retiary.org/ls/music/realaudio/ob_sys/05_alles_synth_improv.rm) 26 July 2011 at the Wayback Machine (info (http://retiary.org/ls/obsolete_systems/) Archived (https://web.archive.org/web/20110824070457/http://retiary.org/ls/obsolete_systems/) 24 August 2011 at the Wayback Machine)
1979[80]	1979[80]	New England Digital	Synclavier II	A commercial digital synthesizer that enabled development of timbre over time by smooth cross-fades between waveforms generated by additive synthesis.	Jon Appleton - Sashasonjon ⓘ
	1996[81]	Kawai	K5000	A commercial digital synthesizer workstation capable of polyphonic, digital additive synthesis of up to 128 sinusodial waves, as well as combing PCM waves.[82]	

Discrete-time equations

In digital implementations of additive synthesis, discrete-time equations are used in place of the continuous-time synthesis equations. A notational convention for discrete-time signals uses brackets i.e. $y[n]$ and the argument n can only be integer values. If the continuous-time synthesis output $y(t)$ is expected to be sufficiently bandlimited; below half the sampling rate or $f_s/2$, it suffices to directly sample the continuous-time expression to get the discrete synthesis equation. The continuous synthesis output can later be reconstructed from the samples using a digital-to-analog converter. The sampling period is $T = 1/f_s$.

Beginning with (**3**),

$$y(t) = \sum_{k=1}^{K} r_k(t) \cos\left(2\pi \int_0^t f_k(u)\, du + \phi_k\right)$$

and sampling at discrete times $t = nT = n/f_s$ results in

$$
\begin{aligned}
y[n] = y(nT) &= \sum_{k=1}^{K} r_k(nT)\cos\left(2\pi\int_0^{nT} f_k(u)\,du + \phi_k\right) \\
&= \sum_{k=1}^{K} r_k(nT)\cos\left(2\pi\sum_{i=1}^{n}\int_{(i-1)T}^{iT} f_k(u)\,du + \phi_k\right) \\
&= \sum_{k=1}^{K} r_k(nT)\cos\left(2\pi\sum_{i=1}^{n}(Tf_k[i]) + \phi_k\right) \\
&= \sum_{k=1}^{K} r_k[n]\cos\left(\frac{2\pi}{f_\mathrm{s}}\sum_{i=1}^{n} f_k[i] + \phi_k\right)
\end{aligned}
$$

where

$r_k[n] = r_k(nT)$ is the discrete-time varying amplitude envelope

$f_k[n] = \frac{1}{T}\int_{(n-1)T}^{nT} f_k(t)\,dt$ is the discrete-time backward difference instantaneous frequency.

This is equivalent to

$$
y[n] = \sum_{k=1}^{K} r_k[n]\cos(\theta_k[n])
$$

where

$$
\begin{aligned}
\theta_k[n] &= \frac{2\pi}{f_\mathrm{s}}\sum_{i=1}^{n} f_k[i] + \phi_k \\
&= \theta_k[n-1] + \frac{2\pi}{f_\mathrm{s}} f_k[n]
\end{aligned}
\quad \text{for all } n > 0
$$ [15]

and

$$
\theta_k[0] = \phi_k.
$$

See also

- Frequency modulation synthesis
- Subtractive synthesis
- Speech synthesis
- Harmonic series (music)

References

1. Julius O. Smith III. "Additive Synthesis (Early Sinusoidal Modeling)" (https://ccrma.stanford.edu/~jos/sasp/Additive_Synthesis_Early_Sinusoidal.html). Retrieved 14 January 2012. "The term "additive synthesis" refers to sound being formed by adding together many sinusoidal components"
2. Gordon Reid. "Synth Secrets, Part 14: An Introduction To Additive Synthesis" (http://www.soundonsound.com/sos/jun00/articles/synthsec.htm). *Sound on Sound* (January 2000). Retrieved 14 January 2012.
3. Mottola, Liutaio (31 May 2017). "Table of Musical Notes and Their Frequencies and Wavelengths" (http://www.liutaiomottola.com/formulae/freqtab.htm).

4. "Fundamental Frequency and Harmonics" (http://www.physicsclassroom.com/class/sound/Lesson-4/Fundamental-Frequency-and-Harmonics).
5. Smith III, Julius O.; Serra, Xavier (2005). "Additive Synthesis" (https://ccrma.stanford.edu/~jos/parshl/Additive_Synthesis.html). *PARSHL: An Analysis/Synthesis Program for Non-Harmonic Sounds Based on a Sinusoidal Representation* (https://ccrma.stanford.edu/~jos/parshl/). Proceedings of the International Computer Music Conference (ICMC-87, Tokyo), Computer Music Association, 1987. CCRMA, Department of Music, Stanford University. Retrieved 11 January 2015. (online reprint (https://ccrma.stanford.edu/STANM/stanms/stanm43/stanm43.pdf))
6. Smith III, Julius O. (2011). "Additive Synthesis (Early Sinusoidal Modeling)" (https://ccrma.stanford.edu/~jos/sasp/Additive_Synthesis_Early_Sinusoidal.html). *Spectral Audio Signal Processing* (https://ccrma.stanford.edu/~jos/sasp/). CCRMA, Department of Music, Stanford University. ISBN 978-0-9745607-3-1. Retrieved 9 January 2012.
7. Roads, Curtis (1995). *The Computer Music Tutorial* (https://archive.org/details/computermusictut00road). MIT Press. p. 134 (https://archive.org/details/computermusictut00road/page/n152). ISBN 978-0-262-68082-0.
8. Moore, F. Richard (1995). *Foundations of Computer Music*. Prentice Hall. p. 16. ISBN 978-0-262-68082-0.
9. Roads, Curtis (1995). *The Computer Music Tutorial* (https://archive.org/details/computermusictut00road). MIT Press. pp. 150 (https://archive.org/details/computermusictut00road/page/n168)–153. ISBN 978-0-262-68082-0.
10. Robert Bristow-Johnson (November 1996). "Wavetable Synthesis 101, A Fundamental Perspective" (https://web.archive.org/web/20130615202748/http://musicdsp.org/files/Wavetable-101.pdf) (PDF). Archived from the original (http://www.musicdsp.org/files/Wavetable-101.pdf) (PDF) on 15 June 2013. Retrieved 21 May 2005.
11. Andrew Horner (November 1995). "Wavetable Matching Synthesis of Dynamic Instruments with Genetic Algorithms" (http://www.aes.org/e-lib/browse.cfm?elib=7923). *Journal of the Audio Engineering Society*. **43** (11): 916–931.
12. Julius O. Smith III. "Group Additive Synthesis" (https://ccrma.stanford.edu/~jos/sasp/Group_Additive_Synthesis.html). CCRMA, Stanford University. Archived (https://web.archive.org/web/20110606200135/https://ccrma.stanford.edu/~jos/sasp/Group_Additive_Synthesis.html) from the original on 6 June 2011. Retrieved 12 May 2011.
13. P. Kleczkowski (1989). "Group additive synthesis". *Computer Music Journal*. **13** (1): 12–20. doi:10.2307/3679851 (https://doi.org/10.2307%2F3679851). JSTOR 3679851 (https://www.jstor.org/stable/3679851).
14. B. Eaglestone and S. Oates (1990). "Analytical tools for group additive synthesis" (http://quod.lib.umich.edu/cgi/p/pod/dod-idx?c=icmc;idno=bbp2372.1990.015). *Proceedings of the 1990 International Computer Music Conference, Glasgow*. Computer Music Association.
15. Rodet, X.; Depalle, P. (1992). "Spectral Envelopes and Inverse FFT Synthesis". *Proceedings of the 93rd Audio Engineering Society Convention*. CiteSeerX 10.1.1.43.4818 (https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.43.4818).
16. McAulay, R. J.; Quatieri, T. F. (1988). "Speech Processing Based on a Sinusoidal Model" (https://web.archive.org/web/20120521071601/http://www.ll.mit.edu/publications/journal/pdf/vol01_no2/1.2.3.speechprocessing.pdf) (PDF). *The Lincoln Laboratory Journal*. **1** (2): 153–167. Archived from the original (http://www.ll.mit.edu/publications/journal/pdf/vol01_no2/1.2.3.speechprocessing.pdf) (PDF) on 21 May 2012. Retrieved 9 December 2013.
17. McAulay, R. J.; Quatieri, T. F. (August 1986). "Speech analysis/synthesis based on a sinusoidal representation". *IEEE Transactions on Acoustics, Speech, and Signal Processing*. **34** (4): 744–754. doi:10.1109/TASSP.1986.1164910 (https://doi.org/10.1109%2FTASSP.1986.1164910).
18. "McAulay-Quatieri Method" (http://www.clear.rice.edu/elec301/Projects02/lorisFor/mqmethod2.html).
19. Serra, Xavier (1989). *A System for Sound Analysis/Transformation/Synthesis based on a Deterministic plus Stochastic Decomposition* (http://mtg.upf.edu/node/304) (PhD thesis). Stanford University. Retrieved 13 January 2012.
20. Smith III, Julius O.; Serra, Xavier. "PARSHL: An Analysis/Synthesis Program for Non-Harmonic Sounds Based on a Sinusoidal Representation" (https://ccrma.stanford.edu/~jos/parshl/Additive_Synthesis.html). Retrieved 9 January 2012.

21. Fitz, Kelly (1999). *The Reassigned Bandwidth-Enhanced Method of Additive Synthesis* (PhD thesis). Dept. of Electrical and Computer Engineering, University of Illinois Urbana-Champaign. CiteSeerX 10.1.1.10.1130 (https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.10.1130).
22. SPEAR Sinusoidal Partial Editing Analysis and Resynthesis for Mac OS X, MacOS 9 and Windows (http://www.klingbeil.com/spear/)
23. "Loris Software for Sound Modeling, Morphing, and Manipulation" (https://web.archive.org/web/20120730195624/http://www.hakenaudio.com/Loris/). Archived from the original (http://www.hakenaudio.com/Loris/) on 30 July 2012. Retrieved 13 January 2012.
24. SMSTools application for Windows (http://mtg.upf.edu/technologies/sms)
25. ARSS: The Analysis & Resynthesis Sound Spectrograph (https://arss.sourceforge.net/)
26. Bonada, J.; Celma, O.; Loscos, A.; Ortola, J.; Serra, X.; Yoshioka, Y.; Kayama, H.; Hisaminato, Y.; Kenmochi, H. (2001). "Singing voice synthesis combining Excitation plus Resonance and Sinusoidal plus Residual Models". *Proc. Of ICMC*. CiteSeerX 10.1.1.18.6258 (https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.18.6258). (PDF (http://mtg.upf.edu/files/publications/icmc2001-celma.pdf))
27. Loscos, A. (2007). *Spectral processing of the singing voice* (PhD thesis). Barcelona, Spain: Pompeu Fabra University. hdl:10803/7542 (https://hdl.handle.net/10803%2F7542). (PDF (http://www.tdx.cat/bitstream/handle/10803/7542/talm.pdf?sequence=1)).
See "*Excitation plus resonances voice model*" (p. 51)
28. Loscos 2007, p. 44, "*Spectral peak processing"*
29. Loscos 2007, p. 44, "*Phase locked vocoder*"
30. Bonada, Jordi; Loscos, Alex (2003). "Sample-based singing voice synthesizer by spectral concatenation: 6. Concatenating Samples" (http://mtg.upf.edu/node/322). *Proc. of SMAC 03*: 439–442.
31. Cooper, F. S.; Liberman, A. M.; Borst, J. M. (May 1951). "The interconversion of audible and visible patterns as a basis for research in the perception of speech" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1063363). *Proc. Natl. Acad. Sci. U.S.A*. **37** (5): 318–25. Bibcode:1951PNAS...37..318C (https://ui.adsabs.harvard.edu/abs/1951PNAS...37..318C). doi:10.1073/pnas.37.5.318 (https://doi.org/10.1073%2Fpnas.37.5.318). PMC 1063363 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1063363). PMID 14834156 (https://pubmed.ncbi.nlm.nih.gov/14834156).
32. Remez, R.E.; Rubin, P.E.; Pisoni, D.B.; Carrell, T.D. (1981). "Speech perception without traditional speech cues". *Science*. **212** (4497): 947–950. Bibcode:1981Sci...212..947R (https://ui.adsabs.harvard.edu/abs/1981Sci...212..947R). doi:10.1126/science.7233191 (https://doi.org/10.1126%2Fscience.7233191). PMID 7233191 (https://pubmed.ncbi.nlm.nih.gov/7233191). S2CID 13039853 (https://api.semanticscholar.org/CorpusID:13039853).
33. Rubin, P.E. (1980). "Sinewave Synthesis Instruction Manual (VAX)" (https://web.archive.org/web/20210829131739/http://www.haskins.yale.edu/featured/sws/SWSmanual.pdf) (PDF). *Internal Memorandum*. Haskins Laboratories, New Haven, CT. Archived from the original (http://www.haskins.yale.edu/featured/sws/SWSmanual.pdf) (PDF) on 29 August 2021. Retrieved 27 December 2011.
34. Sagayama, S. [in Japanese]; Itakura, F. (1979), 複合正弦波による音声合成 [Speech Synthesis by Composite Sinusoidal Wave], *Speech Committee of Acoustical Society of Japan* (published October 1979), S79-39
35. Sagayama, S.; Itakura, F. (October 1979). 複合正弦波による簡易な音声合成法 [Simple Speech Synthesis method by Composite Sinusoidal Wave]. *Proceedings of Acoustical Society of Japan, Autumn Meeting*. Vol. 3-2-3. pp. 557–558.
36. Sagayama, S.; Itakura, F. (1986). "Duality theory of composite sinusoidal modeling and linear prediction". *ICASSP '86. IEEE International Conference on Acoustics, Speech, and Signal Processing*. Vol. 11. Acoustics, Speech, and Signal Processing, IEEE International Conference on ICASSP '86. (published April 1986). pp. 1261–1264. doi:10.1109/ICASSP.1986.1168815 (https://doi.org/10.1109%2FICASSP.1986.1168815). S2CID 122814777 (https://api.semanticscholar.org/CorpusID:122814777).

37. Itakura, F. (2004). "Linear Statistical Modeling of Speech and its Applications -- Over 36-year history of LPC --" (https://web.archive.org/web/20220524112852/https://www.icacommission.org/Proceedings/ICA2004Kyoto/pdf/We3.D.pdf) (PDF). *Proceedings of the 18th International Congress on Acoustics (ICA 2004), We3.D, Kyoto, Japan, Apr. 2004*. **3** (published April 2004): III–2077–2082. Archived from the original (http://www.icacommission.org/Proceedings/ICA2004Kyoto/pdf/We3.D.pdf) (PDF) on 24 May 2022. Retrieved 24 October 2014. "6. Composite Sinusoidal Modeling(CSM) In 1975, Itakura proposed the line spectrum representation (LSR) concept and its algorithm to obtain a set of parameters for new speech spectrum representation. Independently from this, Sagayama developed a composite sinusoidal modeling (CSM) concept which is equivalent to LSR but give a quite different formulation, solving algorithm and synthesis scheme. Sagayama clarified the duality of LPC and CSM and provided the unified view covering LPC, PARCOR, LSR, LSP and CSM, CSM is not only a new concept of speech spectrum analysis but also a key idea to understand the linear prediction from a unified point of view. ..."
38. Adrien, Jean-Marie (1991). "The missing link: modal synthesis" (http://dl.acm.org/citation.cfm?id=131158). In Giovanni de Poli; Aldo Piccialli; Curtis Roads (eds.). *Representations of Musical Signals* (https://archive.org/details/representationso0000unse_k7n4/page/269). Cambridge, MA: MIT Press. pp. 269–298 (https://archive.org/details/representationso0000unse_k7n4/page/269). ISBN 978-0-262-04113-3.
39. Morrison, Joseph Derek (IRCAM); Adrien, Jean-Marie (1993). "MOSAIC: A Framework for Modal Synthesis". *Computer Music Journal*. **17** (1): 45–56. doi:10.2307/3680569 (https://doi.org/10.2307%2F3680569). JSTOR 3680569 (https://www.jstor.org/stable/3680569).
40. Bilbao, Stefan (October 2009), "Modal Synthesis" (https://ccrma.stanford.edu/~bilbao/booktop/node14.html), *Numerical Sound Synthesis: Finite Difference Schemes and Simulation in Musical Acoustics*, Chichester, UK: John Wiley and Sons, ISBN 978-0-470-51046-9, "*A different approach, with a long history of use in physical modeling sound synthesis, is based on a frequency-domain, or modal description of vibration of objects of potentially complex geometry. Modal synthesis [1,148], as it is called, is appealing, in that the complex dynamic behaviour of a vibrating object may be decomposed into contributions from a set of modes (the spatial forms of which are eigenfunctions of the particular problem at hand, and are dependent on boundary conditions), each of which oscillates at a single complex frequency. ...*" (See also companion page (http://www2.ph.ed.ac.uk/~sbilbao/nsstop.html))
41. Doel, Kees van den; Pai, Dinesh K. (2003). Greenebaum, K. (ed.). "Modal Synthesis For Vibrating Object" (http://www.cs.ubc.ca/~kvdoel/publications/modalpaper.pdf) (PDF). *Audio Anecdotes*. Natick, MA: AK Peter. "*When a solid object is struck, scraped, or engages in other external interactions, the forces at the contact point causes deformations to propagate through the body, causing its outer surfaces to vibrate and emit sound waves. ... A good physically motivated synthesis model for objects like this is modal synthesis ... where a vibrating object is modeled by a bank of damped harmonic oscillators which are excited by an external stimulus.*"
42. Prestini, Elena (2004) [Rev. ed of: Applicazioni dell'analisi armonica. Milan: Ulrico Hoepli, 1996]. *The Evolution of Applied Harmonic Analysis: Models of the Real World* (https://books.google.com/books?id=fye--TBu4T0C). trans. New York, USA: Birkhäuser Boston. pp. 114–115. ISBN 978-0-8176-4125-2. Retrieved 6 February 2012.
43. Fourier, Jean Baptiste Joseph (1822). *Théorie analytique de la chaleur* (https://archive.org/details/theorieanalytiqu00fourgoog) [*The Analytical Theory of Heat*] (in French). Paris, France: Chez Firmin Didot, père et fils. ISBN 9782876470460.
44. Miller, Dayton Clarence (1926) [1916]. *The Science of Musical Sounds* (https://archive.org/details/scienceofmusical028670mbp). New York: The Macmillan Company. pp. 110 (https://archive.org/details/scienceofmusical028670mbp/page/n127), 244–248.
45. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*. **49**. Taylor & Francis: 490. 1875. {{cite journal}}: Missing or empty |title= (help)
46. Thomson, Sir W. (1878). "Harmonic analyzer" (https://zenodo.org/record/1432065). *Proceedings of the Royal Society of London*. **27** (185–189). Taylor and Francis: 371–373. doi:10.1098/rspl.1878.0062 (https://doi.org/10.1098%2Frspl.1878.0062). JSTOR 113690 (https://www.jstor.org/stable/113690).
47. Cahan, David (1993). Cahan, David (ed.). *Hermann von Helmholtz and the foundations of nineteenth-century science* (https://books.google.com/books?id=lfdJNRgzKyUC). Berkeley and Los Angeles, USA: University of California Press. pp. 110–114, 285–286. ISBN 978-0-520-08334-9.

48. Helmholtz, von, Hermann (1863). *Die Lehre von den Tonempfindungen als physiologische Grundlage für die Theorie der Musik* (http://vlp.mpiwg-berlin.mpg.de/library/data/lit3483/index_html?pn=1&ws=1.5) [*On the sensations of tone as a physiological basis for the theory of music*] (in German) (1st ed.). Leipzig: Leopold Voss. pp. v.
49. Christensen, Thomas Street (2002). *The Cambridge History of Western Music* (https://books.google.com/books?id=ioa9uW2t7AQC). Cambridge, United Kingdom: Cambridge University Press. pp. 251, 258. ISBN 978-0-521-62371-1.
50. von Helmholtz, Hermann (1875). *On the sensations of tone as a physiological basis for the theory of music* (https://archive.org/details/onsensationston00helmgoog). London, United Kingdom: Longmans, Green, and co. pp. xii, 175–179.
51. Russell, George Oscar (1936). *Year book - Carnegie Institution of Washington (1936)* (https://archive.org/details/yearbookcarne35193536carn). Carnegie Institution of Washington: Year Book. Vol. 35. Washington: Carnegie Institution of Washington. pp. 359 (https://archive.org/details/yearbookcarne35193536carn/page/359)–363.
52. Lodge, John E. (April 1938). Brown, Raymond J. (ed.). "Odd Laboratory Tests Show Us How We Speak: Using X Rays, Fast Movie Cameras, and Cathode-Ray Tubes, Scientists Are Learning New Facts About the Human Voice and Developing Teaching Methods To Make Us Better Talkers" (https://books.google.com/books?id=wigDAAAAMBAJ&pg=PA32). *Popular Science Monthly*. **132** (4). New York, USA: Popular Science Publishing: 32–33.
53. Comerford, P. (1993). "Simulating an Organ with Additive Synthesis". *Computer Music Journal*. **17** (2): 55–65. doi:10.2307/3680869 (https://doi.org/10.2307%2F3680869). JSTOR 3680869 (https://www.jstor.org/stable/3680869).
54. "Institute News and Radio Notes". *Proceedings of the IRE*. **28** (10): 487–494. 1940. doi:10.1109/JRPROC.1940.228904 (https://doi.org/10.1109%2FJRPROC.1940.228904).
55. Douglas, A. (1948). "Electrotonic Music". *Proceedings of the Royal Musical Association*. **75**: 1–12. doi:10.1093/jrma/75.1.1 (https://doi.org/10.1093%2Fjrma%2F75.1.1).
56. Douglas, Alan Lockhart Monteith (1957). *The Electrical Production of Music* (https://archive.org/details/electricalproduc00doug). London, UK: Macdonald. pp. 140 (https://archive.org/details/electricalproduc00doug/page/140), 142.
57. Pejrolo, Andrea; DeRosa, Rich (2007). *Acoustic and MIDI orchestration for the contemporary composer*. Oxford, UK: Elsevier. pp. 53–54.
58. Weidenaar, Reynold (1995). *Magic Music from the Telharmonium* (https://archive.org/details/bub_gb_Gr2kq-598-YC). Lanham, MD: Scarecrow Press. ISBN 978-0-8108-2692-2.
59. Moog, Robert A. (October–November 1977). "Electronic Music". *Journal of the Audio Engineering Society*. **25** (10/11): 856.
60. Olsen, Harvey (14 December 2011). Brown, Darren T. (ed.). "Leslie Speakers and Hammond organs: Rumors, Myths, Facts, and Lore" (https://web.archive.org/web/20120901005950/http://hammond-organ.com/History/hammond_lore.htm). *The Hammond Zone*. Hammond Organ in the U.K. Archived from the original (http://www.hammond-organ.com/History/hammond_lore.htm) on 1 September 2012. Retrieved 20 January 2012.
61. Holzer, Derek (22 February 2010). "A brief history of optical synthesis" (http://www.umatic.nl/tonewheels_historical.html). Retrieved 13 January 2012.
62. Vail, Mark (1 November 2002). "Eugeniy Murzin's ANS – Additive Russian synthesizer". *Keyboard Magazine*. p. 120.
63. Young, Gayle. "Oscillator Bank (1959)" (http://www.hughlecaine.com/en/oscbank.html).
64. Young, Gayle. "Spectrogram (1959)" (http://www.hughlecaine.com/en/spectro.html).
65. Luce, David Alan (1963). *Physical correlates of nonpercussive musical instrument tones* (Thesis thesis). Cambridge, Massachusetts, U.S.A.: Massachusetts Institute of Technology. hdl:1721.1/27450 (https://hdl.handle.net/1721.1%2F27450).
66. Beauchamp, James (17 November 2009). "The Harmonic Tone Generator: One of the First Analog Voltage-Controlled Synthesizers" (https://web.archive.org/web/20180612113043/http://cmp.music.illinois.edu/beaucham/htg.html). *Prof. James W. Beauchamp Home Page*. Archived from the original (http://cmp.music.illinois.edu/beaucham/htg.html) on 12 June 2018. Retrieved 29 May 2018.
67. Beauchamp, James W. (October 1966). "Additive Synthesis of Harmonic Musical Tones" (http://www.aes.org/e-lib/browse.cfm?elib=1129). *Journal of the Audio Engineering Society*. **14** (4): 332–342.

68. "RMI Harmonic Synthesizer" (https://web.archive.org/web/20110609205852/http://www.synthmuseum.com/rmi/rmihar01.html). Synthmuseum.com. Archived from the original on 9 June 2011. Retrieved 12 May 2011.
69. Reid, Gordon. "PROG SPAWN! The Rise And Fall of Rocky Mount Instruments (Retro)" (https://web.archive.org/web/20111225162843/http://www.soundonsound.com/sos/dec01/articles/retrozone1201.asp). *Sound on Sound* (December 2001). Archived from the original (http://www.soundonsound.com/sos/dec01/articles/retrozone1201.asp) on 25 December 2011. Retrieved 22 January 2012.
70. Flint, Tom. "Jean Michel Jarre: 30 Years of Oxygene" (http://www.soundonsound.com/sos/feb08/articles/jmjarre.htm). *Sound on Sound* (February 2008). Retrieved 22 January 2012.
71. "Allen Organ Company" (http://www.fundinguniverse.com/company-histories/Allen-Organ-company-company-History.html). *fundinguniverse.com*.
72. Cosimi, Enrico (20 May 2009). "EMS Story - Prima Parte" (https://web.archive.org/web/20090522022413/http://audio.accordo.it/articles/2009/05/23828/ems-story-prima-parte.html) [EMS Story - Part One]. *Audio Accordo.it* (in Italian). Archived from the original (http://audio.accordo.it/articles/2009/05/23828/ems-story-prima-parte.html) on 22 May 2009. Retrieved 21 January 2012.
73. Hinton, Graham (2002). "EMS: The Inside Story" (https://web.archive.org/web/20130521015858/http://www.ems-synthi.demon.co.uk/emsstory.html). Electronic Music Studios (Cornwall). Archived from the original (http://www.ems-synthi.demon.co.uk/emsstory.html) on 21 May 2013.
74. *The New Sound of Music* (TV). UK: BBC. 1979. Includes a demonstration of DOB and AFB.
75. Leete, Norm. "Fairlight Computer – Musical Instrument (Retro)" (http://www.soundonsound.com/sos/apr99/articles/fairlight.htm). *Sound on Sound* (April 1999). Retrieved 29 January 2012.
76. Twyman, John (1 November 2004). *(inter)facing the music: The history of the Fairlight Computer Musical Instrument* (https://web.archive.org/web/20120323200645/http://www.geosci.usyd.edu.au/users/john/thesis/thesis_web.pdf) (PDF) (Bachelor of Science (Honours) thesis). Unit for the History and Philosophy of Science, University of Sydney. Archived from the original (http://www.geosci.usyd.edu.au/users/john/thesis/thesis_web.pdf) (PDF) on 23 March 2012. Retrieved 29 January 2012.
77. Street, Rita (8 November 2000). "Fairlight: A 25-year long fairytale" (https://web.archive.org/web/20031008201831/http://www.audiomedia.com/archive/features/uk-1000/uk-1000-fairlight/uk-1000-fairlight.htm). *Audio Media magazine*. IMAS Publishing UK. Archived from the original (http://www.audiomedia.com/archive/features/uk-1000/uk-1000-fairlight/uk-1000-fairlight.htm) on 8 October 2003. Retrieved 29 January 2012.
78. "Computer Music Journal" (http://egrefin.free.fr/images/Fairlight/CMJfall78.jpg) (JPG). 1978. Retrieved 29 January 2012.
79. Leider, Colby (2004). "The Development of the Modern DAW". *Digital Audio Workstation*. McGraw-Hill. p. 58.
80. Joel, Chadabe (1997). *Electric Sound* (http://www.pearsonhighered.com/educator/product/Electric-Sound-The-Past-and-Promise-of-Electronic-Music/9780133032314.page). Upper Saddle River, N.J., U.S.A.: Prentice Hall. pp. 177–178, 186. ISBN 978-0-13-303231-4.
81. "Kawai K5000 | Vintage Synth Explorer" (https://www.vintagesynth.com/kawai/k5000). *www.vintagesynth.com*. Retrieved 21 January 2024.
82. "Kawai K5000R & K5000S" (https://www.soundonsound.com/reviews/kawai-k5000r-k5000s). *www.soundonsound.com*. Retrieved 21 January 2024.

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- Digital Keyboards Synergy (http://users.ece.gatech.edu/lanterma/synergy/)

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